

DBMS Question Paper – 50 Marks

Section A – Short Answer Questions

1. Define DBMS and explain its advantages over the file system. (3 Marks)
2. Differentiate between Primary Key and Foreign Key with examples. (3 Marks)
3. Explain the concept of Normalization and its importance in database design. (3 Marks)
4. What is an ER Model? Explain any two components of an ER diagram. (3 Marks)
5. Define Transaction in DBMS. Explain the ACID properties. (3 Marks)

Section B – Medium Answer Questions

6. Explain the three-schema architecture of DBMS. (5 Marks)
7. Describe the different types of joins in SQL with examples. (5 Marks)
8. Explain Functional Dependency and its role in normalization. (5 Marks)
9. Discuss the differences between DDL, DML, and DCL commands with examples. (5 Marks)
10. Explain indexing in DBMS and its advantages. (5 Marks)

Section C – Technical Query Question

11. Consider the following table:

StudentID	Name	Department	Marks
101	Ravi	CSE	85
102	Anu	ECE	78
103	Kiran	CSE	92
104	Meena	EEE	88

- a) Display the names of students who belong to the CSE department. (3 Marks)
- b) Find the average marks of all students. (3 Marks)
- c) Display the student details in descending order of marks. (4 Marks)

```
-- a)
SELECT Name
FROM Student
WHERE Department = 'CSE';
```

```
-- b)
SELECT AVG(Marks) AS Average_Marks
FROM Student;
```

```
-- c)
SELECT *
FROM Student
ORDER BY Marks DESC;
```